Wi-Fi Performance Analysis

Automated network performance monitoring across multiple locations, designed for large spaces like airports where reliable Wi-Fi is critical. This project collects, analyzes, and visualizes network metrics to ensure consistent connectivity despite challenges like user density and interference.

By:

Team – 3

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Problem Statement: Wi-Fi Challenges in Large Spaces

Performance Variability

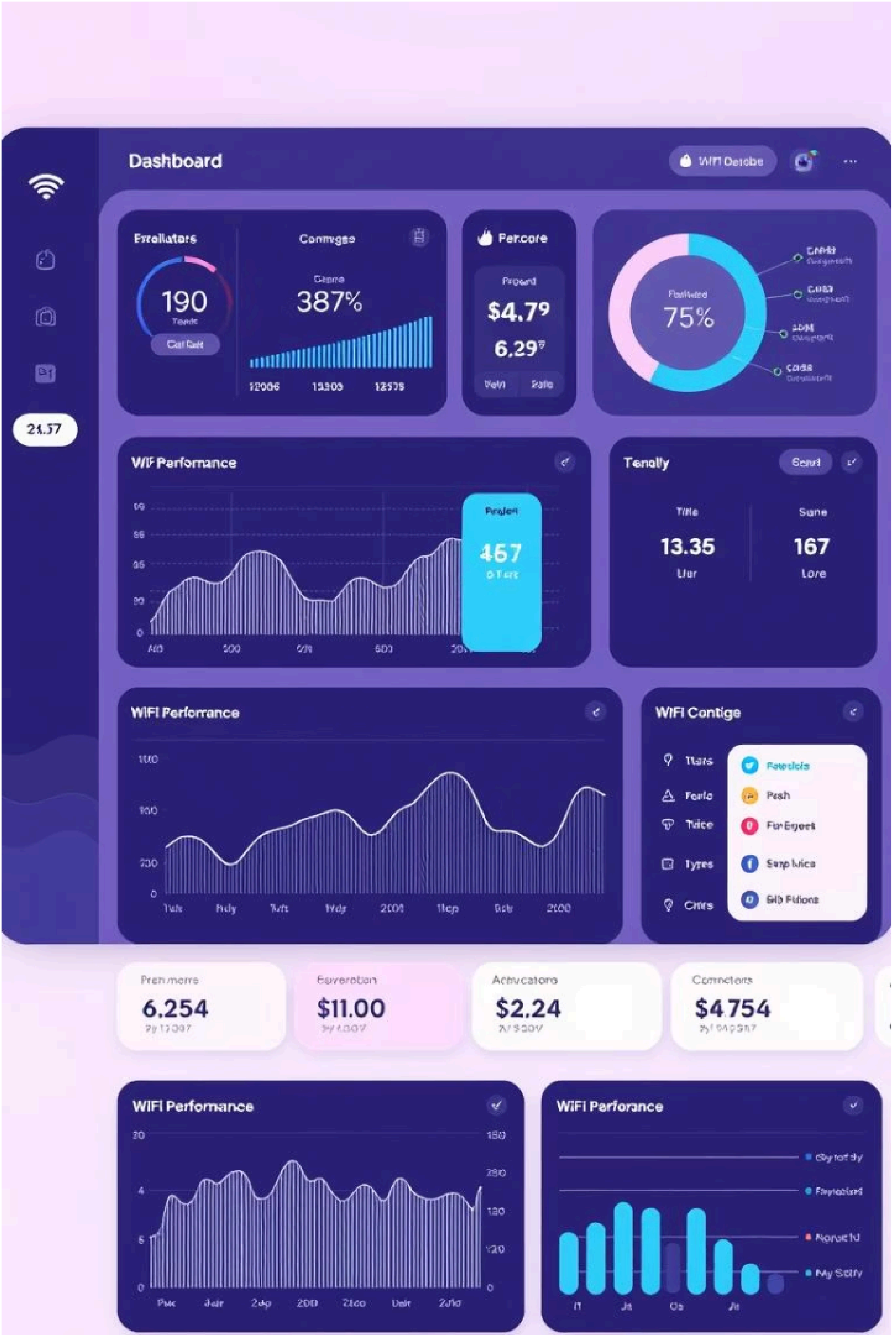
Wi-Fi fluctuates during peak hours due to high user density and physical obstructions.

Interference Issues

Multiple networks and devices cause signal interference, degrading connectivity.

No Centralized View

Existing tools are either too technical or fragmented, lacking a unified dashboard for quick and clear WiFi performance assessment.



Project Objectives

Automated Monitoring

Simple user interface to trigger continuous Wi-Fi data collection.

Comprehensive Metrics

Collect download/upload speeds, RSSI, latency, jitter, and packet loss.

Insightful Visualization

Trend analysis across time and locations to support network optimization.

Use Case: Airport Wi-Fi Monitoring

High User Demand

Thousands of passengers rely on public Wi-Fi for communication and entertainment.

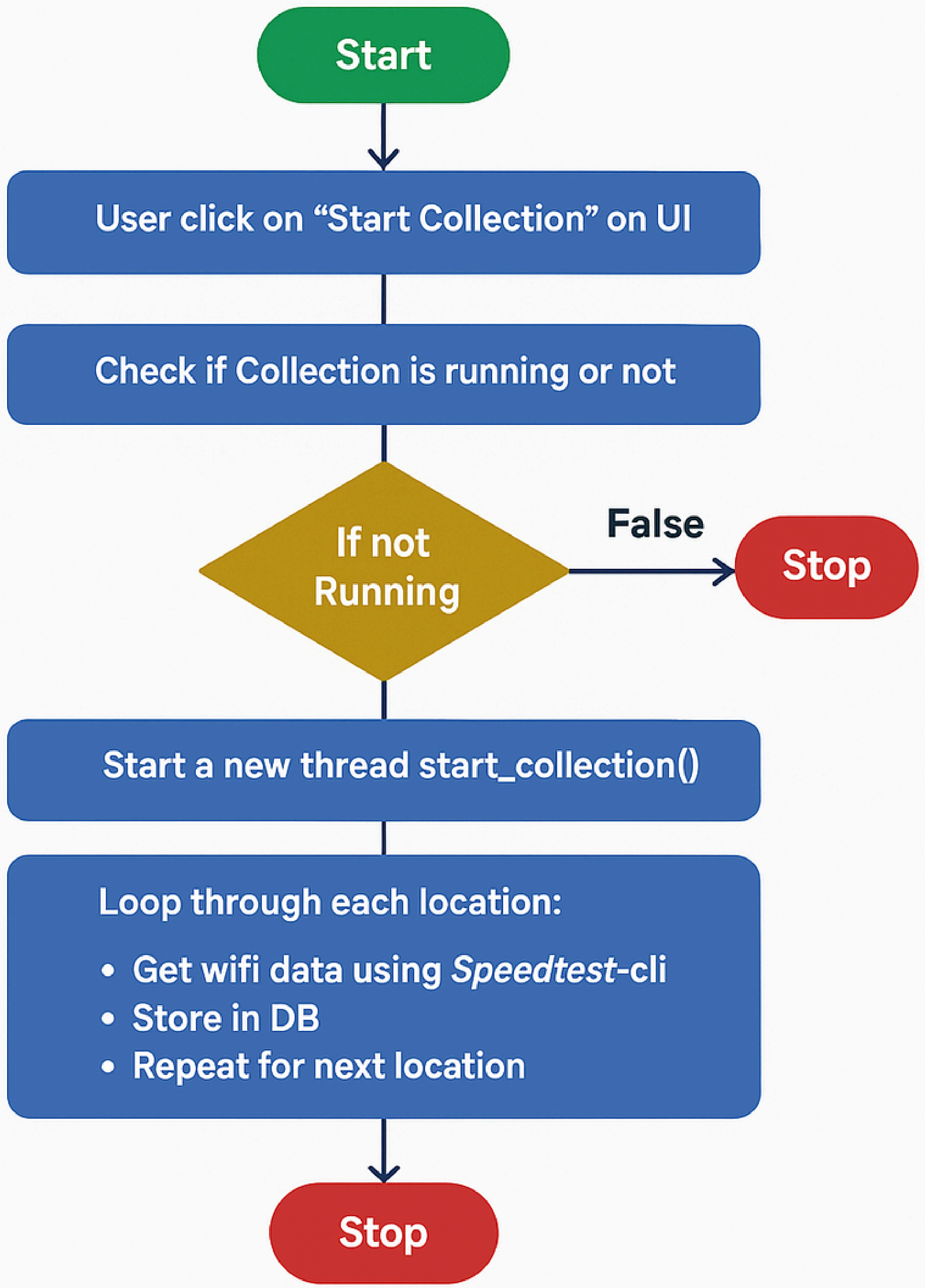
Operational Needs

Airport staff require stable connectivity for ticketing, security, and logistics.

Coverage Challenges

Large areas with varying signal strength demand consistent Quality of Service.

Data Collection Part



Implementation Workflow

1

Data Collection

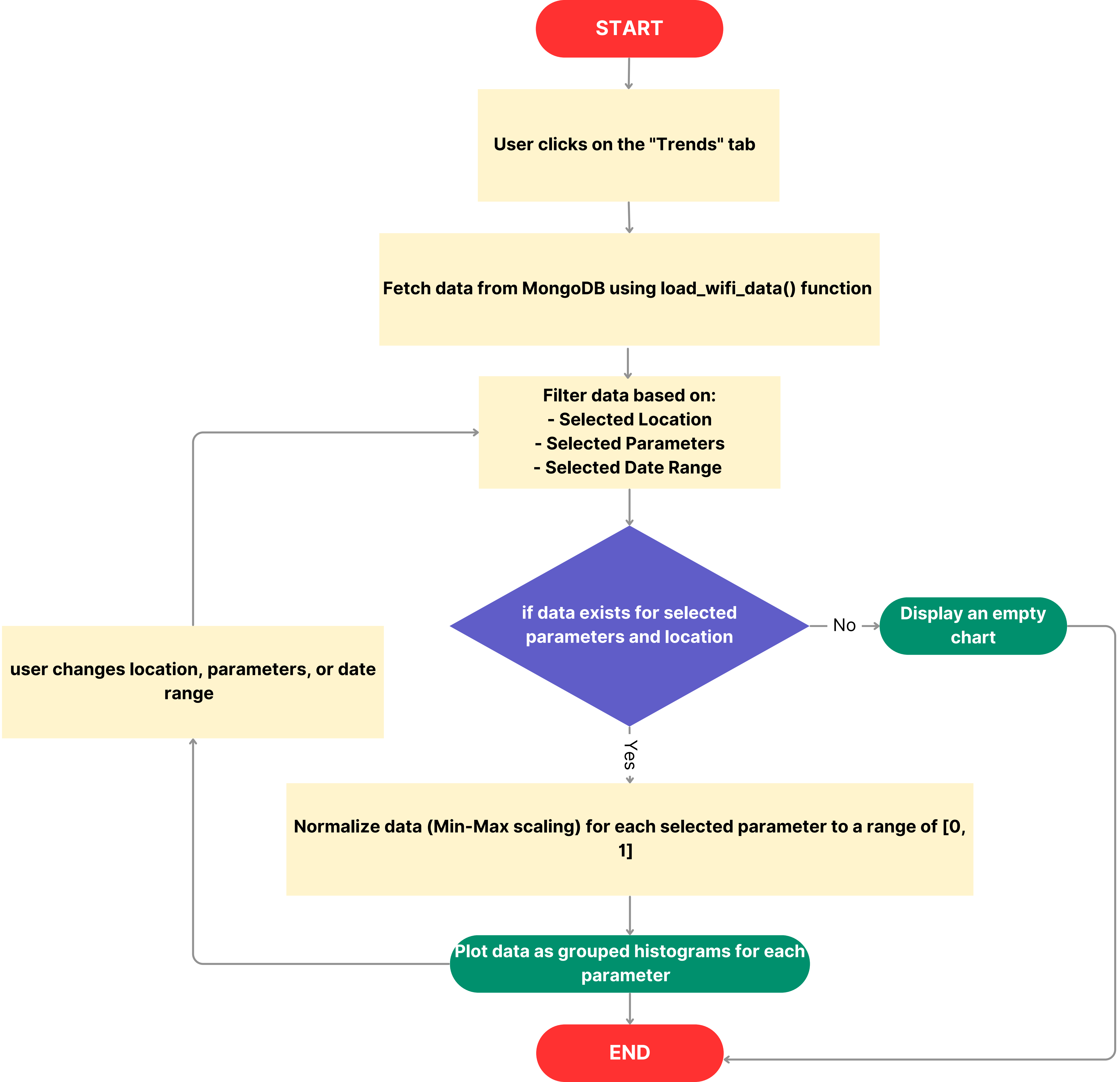
User triggers collection → Threaded scans across locations → Metrics stored in database.

2

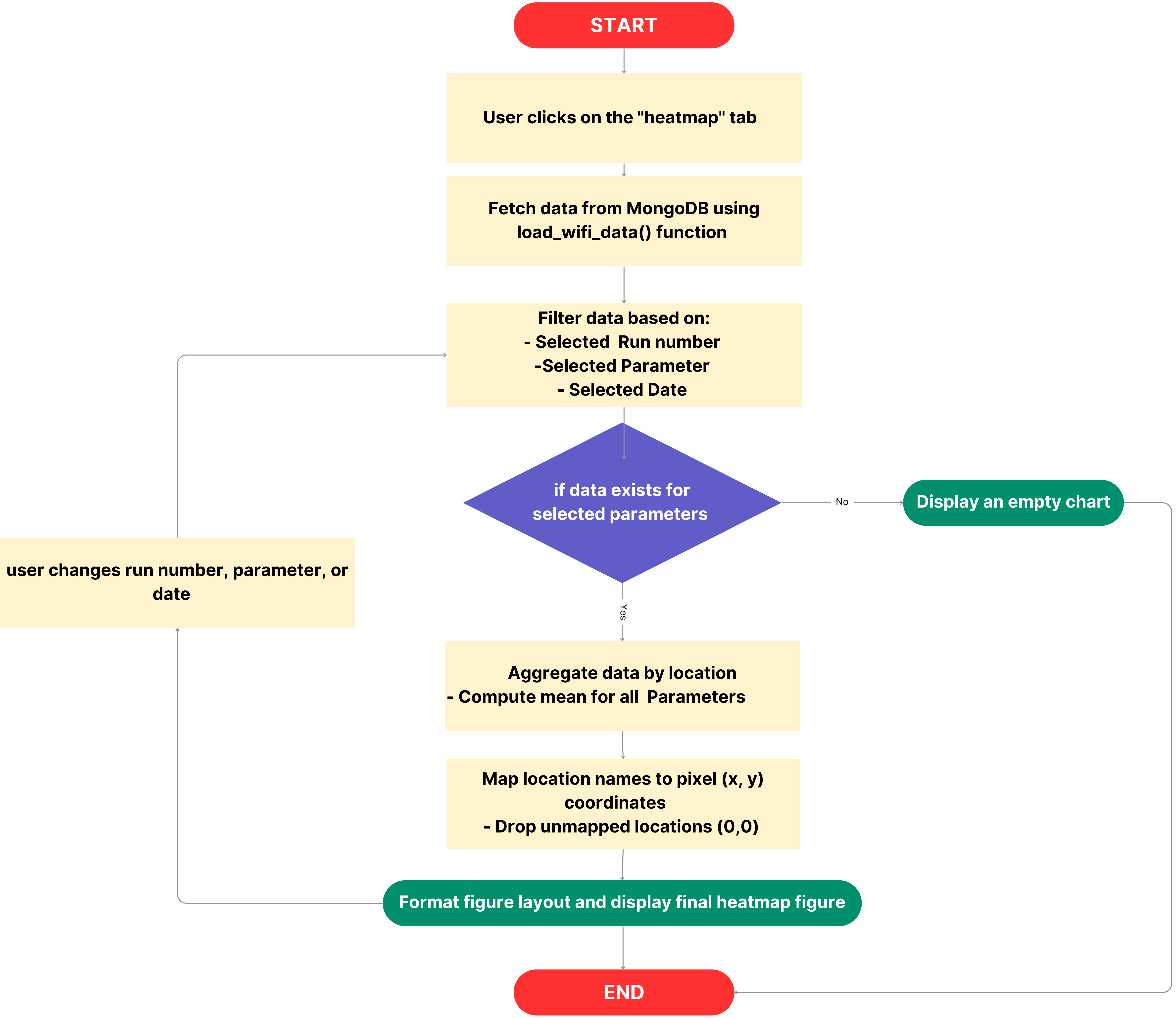
Data Analysis

Data fetched into DataFrames → Overview and run-specific analysis → Trend and heatmap visualizations.

Workflow for Trends tab



Workflow for Heatmaps tab



Key Features of the

Real-time Monitoring

Continuous data collection with minimal manual input.

Location Analysis

Performance evaluated across multiple physical locations.

Trend Visualization

Track network performance over time for informed decisions.

Heatmaps

Visual comparison of signal strength and quality by location.

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Development Challenges & Solutions

Handling Missing Data

Filtered zero or missing values with clear labeling to maintain data integrity.

Execution Consistency

Implemented threading to ensure reliable and continuous scans.

Visualization Issues

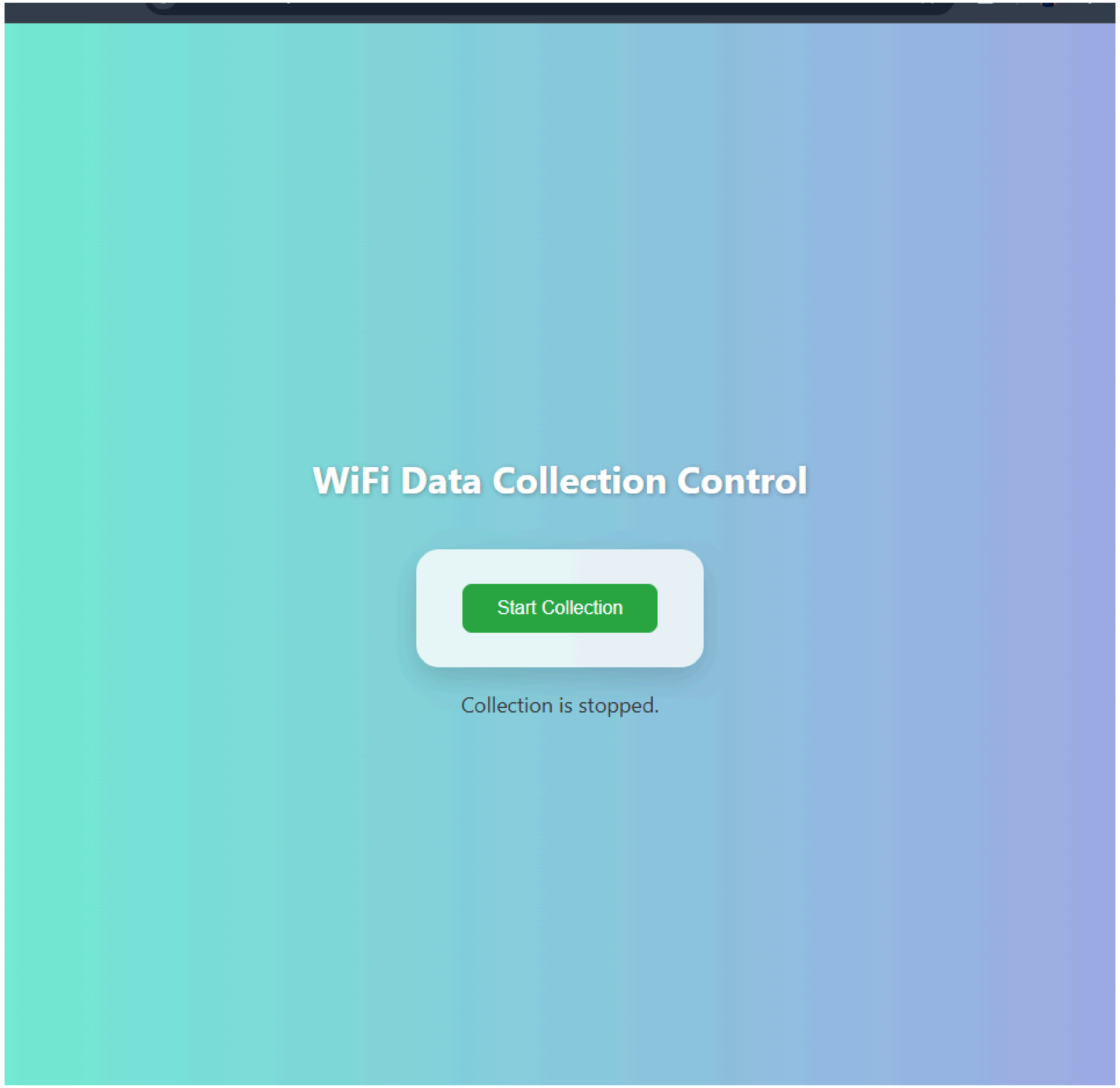
Normalized mixed-scale parameters and refined legends for clarity.

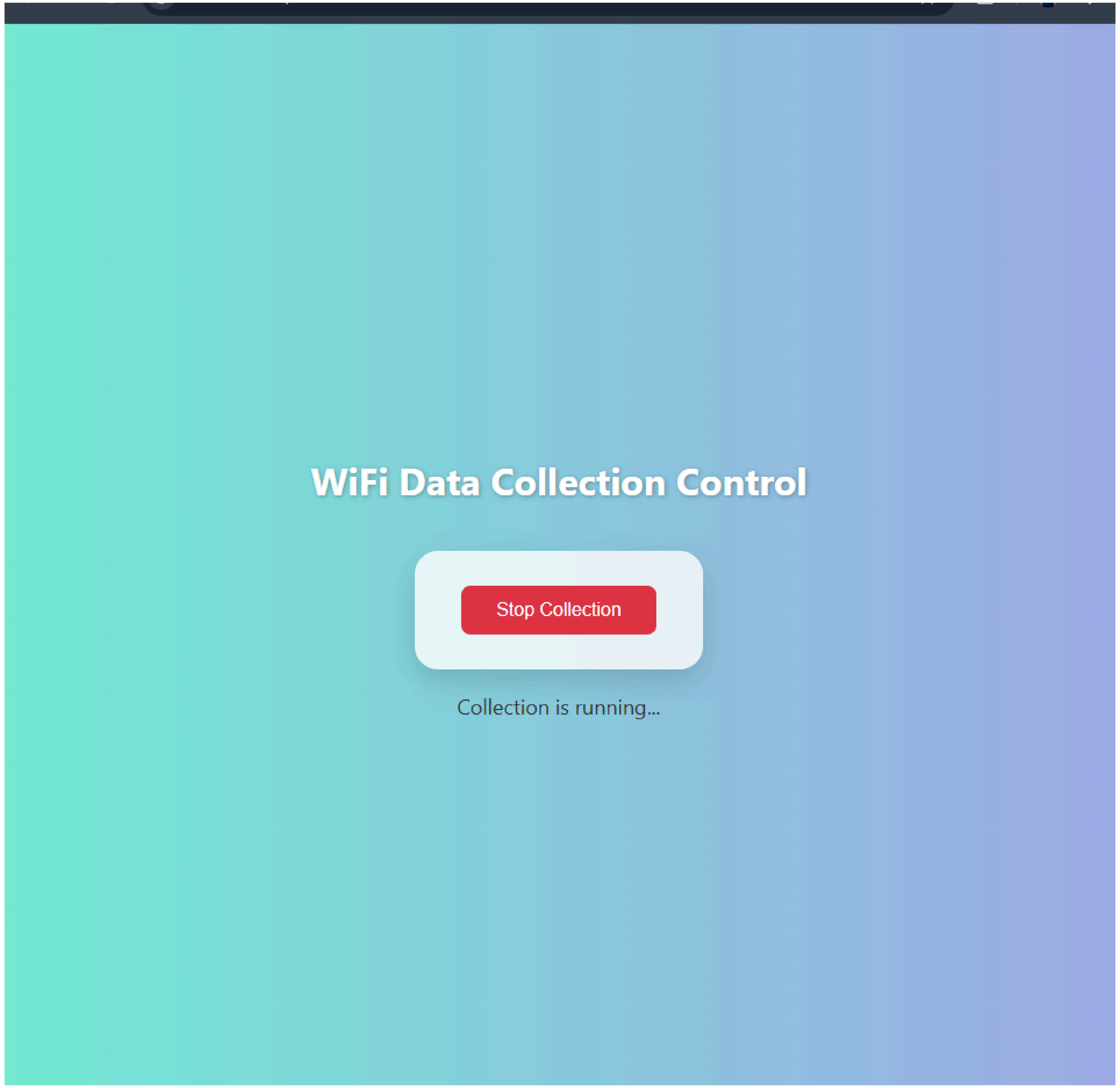
UI Efficiency

Improved layout and responsiveness for better user experience.

Challenges Faced	Solutions & Workarounds
Zero or missing values (e.g., upload speed = 0)	Filtered out null/zero entries and labeled them clearly during visualization
Visualization of sparse or inconsistent data	Simulated multiple test runs to generate more consistent and analyzable data
Grouped charts with mixed-scale parameters	Applied normalized values, and added clear legends for better readability
Choosing appropriate visualization types	Experimented with various chart styles → finalized on **bar graphs** and **heatmaps**
UI inefficiency during early stages	Refined layout for better toggle controls, and responsiveness

Data collection - UI





Results &

Automated Monitoring

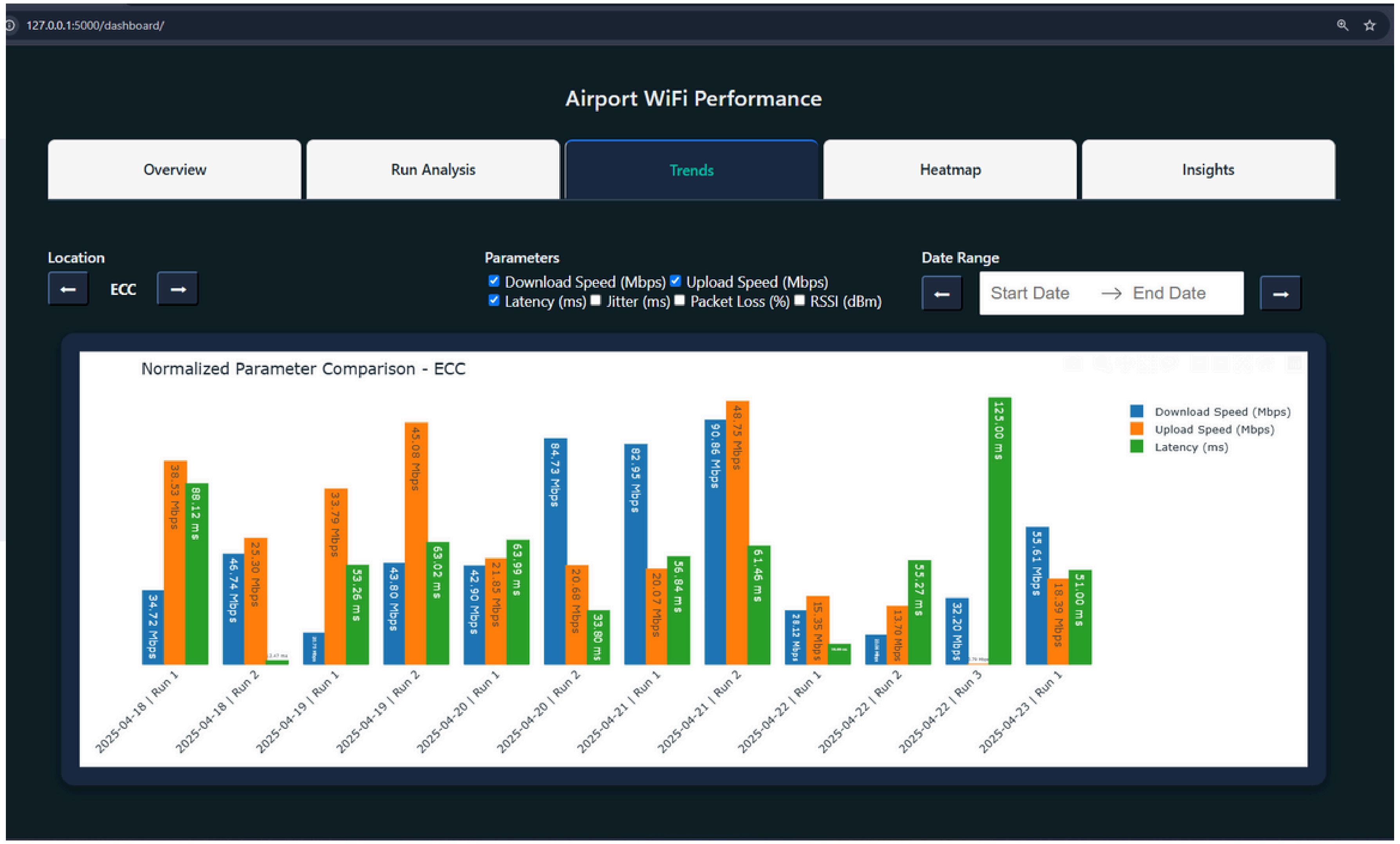
Reduced manual effort with continuous, automated data collection.

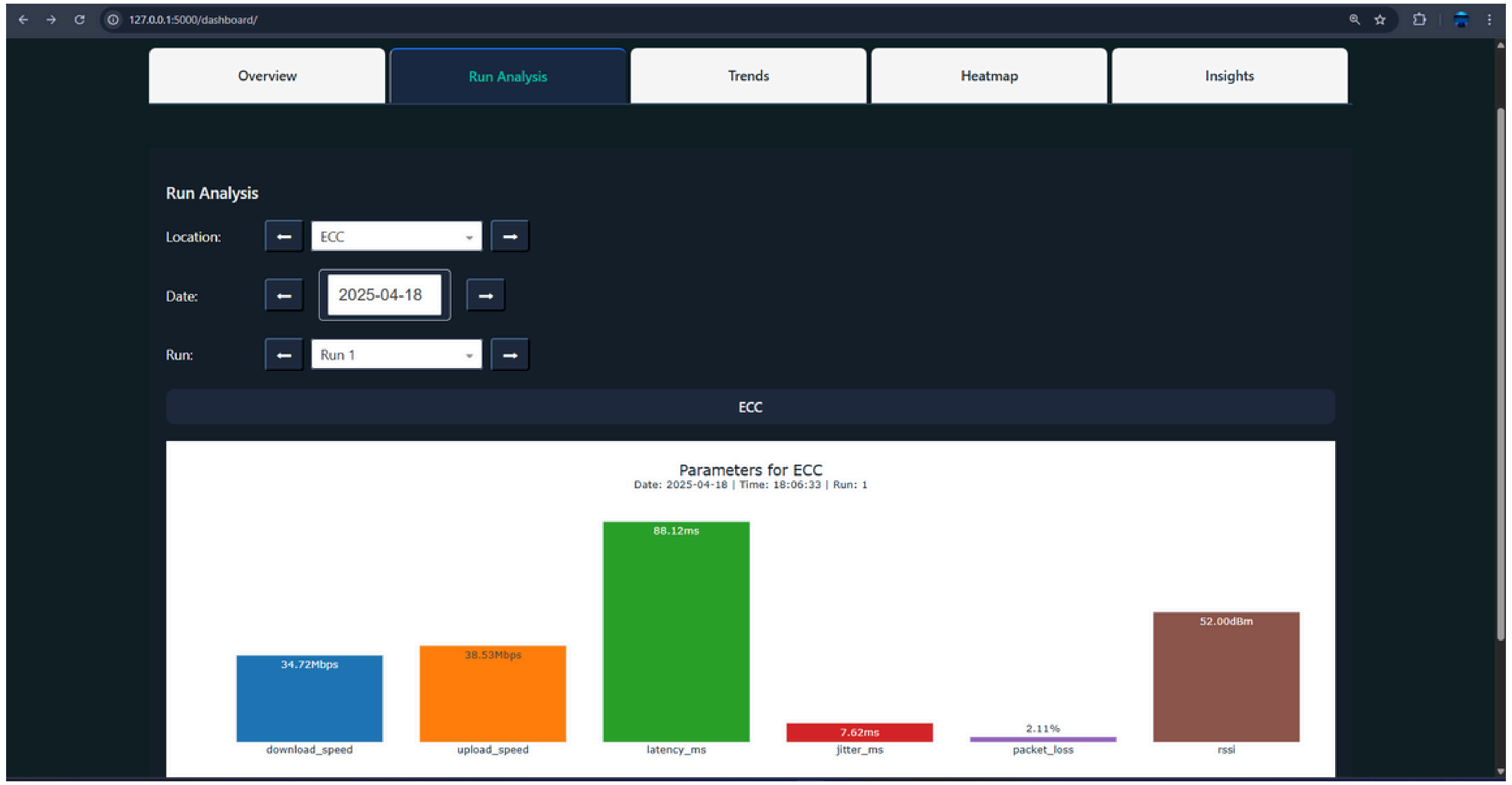
Real-time Insights

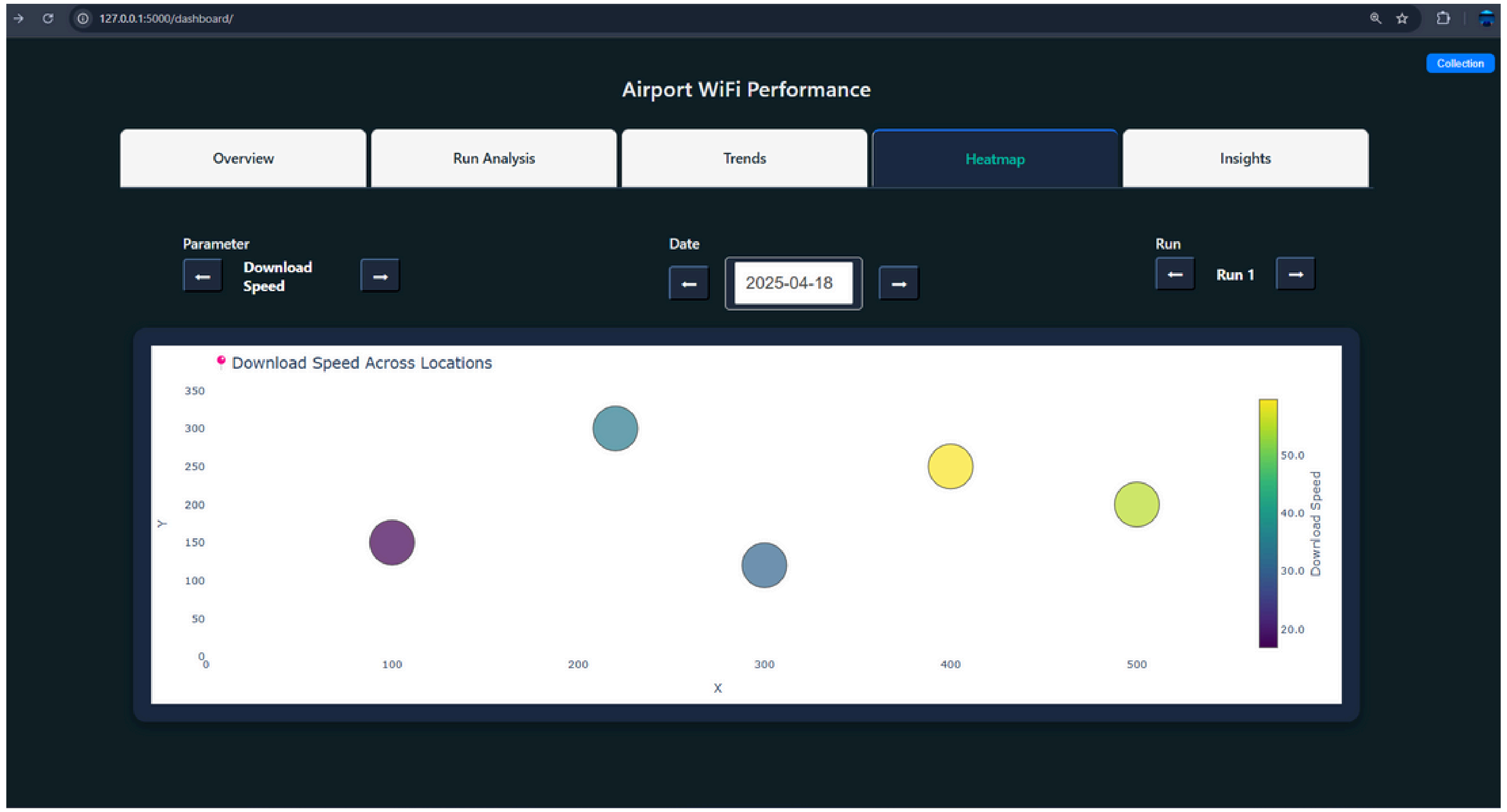
Efficient storage and clear visual analysis of network performance.

Scalable Design

Modular architecture adaptable to various large environments







Future Enhancements

Expanded Metrics

Include browsing speed and streaming quality for deeper insights.

Real-Time Dashboard

Auto-refreshing interface with smart alerts for threshold breaches.

Cross-Platform Support

Mobile compatibility and automated optimization suggestions.

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Thank you